

Aviation Cybersecurity Final Report

Nikolay Boiarski

Supervisor: Olaf Maennel

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1 Abstract

This project focused on cybersecurity in aviation. With the current rise of new technologies in aviation, specifically UAVs, the need for secure communication becomes more and more apparent as current communication methods have little to no security measures in place. This project focused on this gap of secure communication by implementing Transport Layer Security And Public Key Infrastructure methods into a virtualised environment to create a baseline system for secure communication. This project found that implementing these systems can dramatically increase security in this sector.

2 Introduction and Motivation

Aviation is one of the largest sectors in the whole world bringing in hundreds of billions of dollars even after the coronavirus pandemic which greatly affected the whole industry Statistica (2021). Typically, the aviation industry consists of manned crews on board the aircraft, but with advances in technology the usage of unmanned aerial vehicles is increasing with new companies such as Volocopter developing drones for multiple tasks such as drones for logistics and air taxi services Volocopter (2023) and large companies such as Amazon developing services such as Prime Air for drone deliveries Amazon (2022). However, while the technology is being developed to make these dreams a reality, there is no focus on making sure these technologies are safe from cyber threats, more specifically threats related to communication between all relevant systems in aviation.

There is existing research on how a network of

drone would look like such as the work by Gharibi et al. (2016) as well as some research for a traffic management system similar to how air traffic control systems currently operate to enable drone operations close to airports and for recreational and commercial use FAA (2023d). This system could operate similar to manned aircraft with drone operators asking for authorisation and giving a flight plan to an automated drone traffic management system which would make sure the drone is operating safely, avoiding collisions, etc. with communications occurring over the ground rather than through the drone as both the drone operator and traffic management system are ground based, however, this system fails to address how messages would be sent between drones, drone operators and traffic management systems securely to make sure malicious actors cannot disrupt normal operations and cause dangerous accidents.

3 Literature Review

To create a secure drone traffic management system, first a review has to be done on any existing research on such systems. Second, an analysis on potential threats in aviation communication and on drones must be done to understand what parts of the system need to be secured and finally any literature for solutions to these security risks must be analysed.

3.1 Existing Drone Traffic Management Solutions

When looking to implement a drone traffic management system with a focus on cybersecurity, it is important to look at existing ideas for such a system

where one can be found from the Federal Aviation Administration (FAA). One of the focuses of the FAA is an air traffic management system with a recent focus on an Unmanned Aircraft System Traffic Management (UTM) FAA (2023c). The UTM project works by having communications operator to operator and operator to the FAA where communications happen through a distributed network rather than a voice channel. This allows real time communication between the FAA and operators and between operators themselves where it is up to them to make sure their operations are safe. The UTM project also has a focus on security where the FAA and multiple aviation communities are collaborating with the International Civil Aviation Organisation (ICAO) to "define a cybersecurity network and identity policies for the International Aviation Trust Framework" (IATF) FAA (2023a). However, this framework focuses on creating a secure network before operating a drone such as what the operator of a drone must do to register it, what information is stored and where and how that generates a certificate that can be used to validate the drones registration and operator license when initially contacting the traffic management system. This framework does not focus on how secure communications would work during drone operations.

The European Union Aviation Safety Agency (EASA) is working on the Urban Air Mobility program which involves the usage of aircraft for transportation of passengers and cargo EASA (2023) where eventually drones could be used instead of pilots. This program does not look at how communication between these aircraft would work and how this communication would be made secure against cyber attacks.

Another project looking into drone traffic management is the U-space project which is an airspace within a controlled airspace which can be reconfigured as required. This allows air traffic control (ATC) to limit areas of the airspace to accommodate for manned aircraft. However, U-space is not managed by ATC and instead by service providers which must provide identification, geo-awareness, flight authorisation and traffic information services SKYbrary (2021). This system does mention how it will make sure messages sent between operators and controllers

will be secure.

3.2 Manned Aviation Cyber Risks

While current research into a drone traffic management systems do not look into the security of these systems, there is a need to analyse what potential cyber risks exist in aviation. For manned aircraft, air to ground communication is one of the most important services while having few protections against potential threats.

3.2.1 VHF Communication

For communication Very-High Frequency (VHF) voice communication systems are used where due to the used frequencies being public and communication not being encrypted, VHF is vulnerable to eavesdropping, spoofing and jamming Elmarady & Rahouma (2021). An eavesdropping attack allows everyone on the same frequency to listen to the messages; A spoofing attack is where unauthorised instructions can be sent to the aircraft or ground station; Jamming attacks cause interference which can block any other messages on the VHF frequency Elmarady & Rahouma (2021).

3.2.2 CPDLC

Another method of communication is Controller-Pilot Data Link Communication where messages can be sent as digital messages which has the benefit of reducing congestion in VHF voice channels. Due to this system being unencrypted, it is susceptible to man in the middle attacks such as eavesdropping, jamming, flooding, injecting incorrect messages, impersonation and others Elmarady & Rahouma (2021).

3.3 Drone Aviation Cyber Risks

While it is important to analyse the threats against manned aircraft, it is also important to analyse these threats against drones to see if there are similarities.

3.3.1 Dos Attacks

A very common type of attack against a drone is a Denial of Service (DoS) attack. This attack works by sending a large amount of requests to the drone which can cause legitimate requests to be ignored which can cause a loss of control of the drone. This type of attack can lead to a loss of identification of the drone, allowing an attacker to completely control the drones movements Hassija et al. (2021).

3.3.2 Man in the Middle Attack

A man in the middle attack allows eavesdropping, jamming, impersonation and other attacks which is typically caused by lack of encryption Hassija et al. (2021).

3.3.3 GPS Spoofing

Drones require signals from GPS satellites for communication which allows for GPS spoofing attacks. These attacks send a controller false information about the path the drone is taking or even forcing the drone to take a different trajectory without the original operator being able to intervene.

3.3.4 Solutions

Hassija et al. (2021) proposes the use of blockchain, Software Defined Networking, Communication using Machine learning and fog computing all combined due to the ability of each method to defend against various attacks against a drone. However, while this research addresses traffic management for automated drones, it does not research cases where there will be a drone operator involved and how communications between this operator and a traffic management system would work.

3.4 Secure Communication Between ATC and Drone Pilots

Apostolovski (2021) proposes a system that uses Transmission Control Protocol/Internet Protocol methods instead of radios and satellites as a communication method between drone operators and a

traffic management system due to both being on the ground allowing ground to ground communications. Apostolovski (2021) proposes a layered encryption scheme with transport layer security (TLS) encryption used as an outside layer and public key infrastructure (PKI) using elliptic curve cryptography (ECC) as an inside layer. The PKI method would work as follows: a drone operator would obtain a valid drone license where upon registration, the operator would get an encrypted certificate to be used for authentication. To communicate with a traffic management system, the drone operator would contact the system which would use the certificate to authenticate the operator Apostolovski (2021). While this method creates a secure connection between a traffic management system and a drone operator, it does not create a secure connection between the operator and the drone.

3.5 Conclusion

In conclusion, there are a few organisations creating drone traffic management systems, however, they are lacking a focus on the security of both communications between operators and the traffic management system. Hassija et al. (2021) addresses the use of multiple methods such as blockchain to secure communications between an operator and a drone while Apostolovski (2021) addresses secure communications between traffic management and operators. However, there are no implementations of these ideas which leads into the plan for this project.

4 Plan

For such a large project the first step is to lay out a plan for what must be done including the minimum requirements, steps to reach the end goals and contingency plans.

The first part of the project will be to make a literature review. Existing literature on security methods in aviation as well as any possible implementation of secure communication methods will be explored to help find the gap in the field.

The next step will be to set up the virtualisation

environment. This environment will be set up on VirtualBox with the Ubuntu Linux operating system. This will be installed on all the virtual machines. The next step to set up the environment involves downloading the PX4 simulation environment to allow for simulation for a drone. If setting up such an environment proves to be too difficult, a simple script that constantly sends data will be used instead as the focus is on secure communication which does not require an actual drone.

After virtualisation, basic communication must be implemented. This will be done using the socket library in Python to allow for the virtual machines to communicate with each other. This will be set up so that the machines send and receive basic data to make sure that this communication works. If this method does not work then other methods of communication between virtual machines on the same network will be explored.

After setting up basic communication, MAVLink will be explored to connect the Python socket script to the PX4 simulation environment which will allow for a constant stream of data from the simulated drone to be sent wherever it is required. The contingency if MAVLink is too difficult to implement will be to use the same basic communication contingency in the virtualisation step.

After getting drone telemetry data sent between the two virtual machines, PKI and TLS methods will be implemented. TLS will be used to encrypt the messages sent across the sockets and certificates will be generated to authorise both virtual machines to send and receive data from each other. If this can not be set up in an appropriate time then the contingency is to explore how insecure the messages are and to explore that malicious users can do with this data.

Once all the communication is set up and made secure, Wireshark will be used to analyse the packets sent between the virtual machines to see what the data looks like when it is encrypted and when it is unencrypted. The PKI implementation will also be tested by having another virtual machine running a drone simulation attempt to connect to the traffic management virtual machine to make sure that it gets rejected. The contingency if this is too difficult to do will be to analyse the python code and show

that it is safe as it is designed in a way to make sure that the data is encrypted with TLS and that PKI is used for authentication.

Lastly, a stretch goal is to attempt to connect two drones to the traffic management system at the same time which will most likely require the socket code to be updated. If this proves too difficult to achieve in a reasonable time frame, it will be ignored as it is a stretch goal for the system and will be left as future work.

5 Methodology

This section will go through the methodology used to achieve secure communication between simulated drones and a traffic management system. The focus was on cybersecurity in aviation, more specifically, secure communication between UAVs which was achieved using drone simulation using Gazebo drone simulation, sockets for data communication and Public Key Infrastructure (PKI) with Transport Layer Security (TLS) encryption for secure communication.

5.1 Virtualisation

To get secure communication between a drone and a traffic management system, the decision to use virtualisation was made due to it being much easier to test security concepts on virtual hardware where everything can be configured as needed as opposed to physical. This also allows for the project to be scaled and replicated in the future on other machines, allowing the results obtained to be used to guide future research in secure communication in the aviation industry.

5.1.1 Virtual Machine and Operating System

The chosen virtual machine software for this project was Oracle VM VirtualBox on a windows based machine. Ubuntu, a Linux based operating system was used on the VirtualBox software due to the software libraries being more optimised for Linux use.

5.1.2 Simulation Software

The basic requirements for this project included a virtual machine to simulate a drone and a virtual machine to simulate a traffic management system. The decision was made to not have a drone controller as communications between a drone and the controller and a controller and a traffic management system can use the same techniques used for communication between a drone and the traffic management system developed in this project. This also reduced the complexity of the project as the focus was on developing a secure communication method rather than building a simulated drone network. The traffic management system made use of sockets to receive data from the simulation virtual machine. while the simulation virtual machine made use of the PX4 autopilot software. It was chosen due to it being part of the Dronecode foundation, which aims to set a standard for open-source drone projects and includes drone simulation and control projects, communication projects and a ground station project. This allows for future work on creating a simulated environment where there is a ground station controlling one or multiple drones communicating with a traffic management system that has all the features required in place which was not in the scope of this project as it focused only on secure communications between these systems. Due to this software being part of the Dronecode foundation, it allowed for PX4 simulated drones to use a communication protocol called MAVLink to communicate with the traffic management system in this project.

5.2 Python Libraries

To achieve the goal of secure communication, some code would be required which was written in the Python programming language due to its ease of use and multiple available libraries as well as integration with other tools specific to drone simulation of which the libraries were 'socket' 'pymavlink' and 'ssl'.

5.2.1 Socket Library

The socket library was used for basic communication between virtual machines, one being a drone simula-

tion and the other a traffic management system. The information between the two virtual machines was the simulated drones altitude, speed and location.

5.2.2 Pymavlink

The pymavlink library was used to use a lightweight messaging protocol for communication with drones called MAVLink (2023). This was used to take real time data from the drone and send it to the traffic management system.

5.2.3 SSL Library

The ssl library was used for both TLS encryption of the messages and for the PKI used in the system.

5.3 PKI and TLS

Using PKI and TLS for secure communication first requires an understanding of the systems.

TLS is a widely used security protocol for privacy and data security for communication systems using the internet Cloudflare (2023). The way it works is by using two encrypted keys, a private key and a public key. The private key is generated first and is used to generate the public key that can be distributed to everyone on a network, in this case a drone to traffic management system communication network. The public key can then be used to encrypt messages that can be only be decrypted by the private key while also being able to decrypt messages encrypted by the private key. This allows for secure communication between the parties with a public and private key pair and due to the private key being difficult to decrypt with the public key, this method is very secure Cockroach-Labs (2023).

While encryption can be useful for secure communication it becomes even more secure with the help of identity authentication which is what PKI provides Cockroach-Labs (2023). Identity authentication is important in this system because if an attacker pretended to be a drone and got access the public key then they could modify the messages causing harm to the whole system. PKI deals with this issue by

having subscribers, relying parties and certificate authorities. A subscriber wants to have a valid identity that can be used to connect to required services. In this context, a subscriber would be a drone and its pilot who wish to prove that they are a valid operator of a drone and can connect to the traffic management system. A relying party wants to use PKI to authenticate and communicate with subscribers and in this context would be the traffic management system for a certain area. The certificate authority (CA) is used to verify the identity of both subscribers and the relying party. In this context there was no official CA and certificates for verification were generated specifically for this project. Ideally a CA would be a governing body for aviation either for a country or one that governs over aviation for all countries, however, this discussion was not in the scope of this project.

5.4 Wireshark

Since the goal of using TLS and PKI is to enable secure communication, a method to verify that there is in fact secure communication must be used. To achieve this a popular network packet analysing tool Wireshark was used due to its popularity and clear documentation making it simple to use Wireshark (2023). For this project Wireshark was used to analyse the packets sent between a simulated drone and a traffic management system to see if the data was encrypted after implementing TLS and PKI methods. While Wireshark is not likely to be a tool used by malicious attackers, it still shows what the messages between the two systems look like which is the information attackers would have access to.

6 Experimental Setup

This section will go through the technical implementation of the tools and methods described in the previous methodology section. It will go through the setup for the virtual machines, PX4 software, and python socket code and the testing methods used to see if the communication was secure or not.

6.1 Virtual Machine Setup

The setup for the virtual machines involved having one to simulate a drone and one to simulate a traffic management system. Both virtual machines were created on VirtualBox with the Ubuntu 20.04 LTS operating system installed. To set up the virtual machines, the process outlined in the official Ubuntu documentation was used Canonical (2023). Each virtual machine was given 35GB of storage and 4GB of RAM to allow for smooth performance and no storage issues when installing various software.

6.2 Drone Simulation and Communication

For drone simulation, PX4 autopilot version 1.12.0 was used. Following the documentation on the PX4 website, the Gazebo simulation software was installed which uses the PX4 software in the loop simulation. This allowed the project to have a simulated drone that could provide real-time telemetry data such as altitude, speed and location.

To allow basic communication between the two virtual machines, the socket and Pymavlink libraries were used. The code for the drone starts by creating a mavlink connection, creating a socket connection and sending telemetry data to the traffic management system. The full code can be seen on the GitHub while the pseudocode is:

```
Create MAVLink connection
Create socket
Connect to traffic management system
Send constant live simulated telemetry
```

The code for the traffic management system involved a similar socket setup while also having a step to listen for any incoming connections. The full code can be seen on the GitHub while the pseudocode is:

```
Create socket
Bind socket to address
Listen for connection
Accept incoming connections
Keep printing received telemetry
```

After creating basic communication, the next step is to encrypt the data with TLS and add PKI verification.

6.3 PKI and TLS Configuration

The goal of this project is to set up secure communication which was done using PKI and TLS. The first step to implement this is to generate the necessary certificates for verification. OpenSSL was used as it contains tools for TLS encryption methods. This project created a Certificate Authority (CA) as well as all the other required certificates and keys for the whole system. To make the CA certificate and a private key the following commands were used:

```
openssl generate RSA CA key
openssl create CA certificate
```

After the CA certificate was made, public keys and certificates were made for both the drone simulation and the traffic management system:

```
openssl generate RSA drone key
openssl generate RSA server key
openssl drone certificate request
openssl server certificate request
openssl accept drone certificate
openssl accept server certificate
```

Due to the CA not being a real trusted CA and instead being generated, to be trusted by the operating system, the CA certificate was moved to where Ubuntu stores trusted certificates. To easier manage the connection and resolve issues with deprecated features from older Python versions, the IP address of the traffic management system was added to the hosts file on the drone simulation virtual machine. The Python code then had to be modified to add a check for the PKI certificates and to wrap the message with PKI encryption.

6.4 Test Methods

6.4.1 Wireshark Testing

To test the TLS encryption was properly implemented and the communication was secure, Wireshark was used to see the packets sent between the

two virtual machines to see what the data looks like with and without encryption. To do this, Wireshark was installed on the traffic management system virtual machine. Wireshark was set to scan on enp0s3 and more specifically, was set to look for data coming to and from the IP address of the traffic management system.

6.4.2 PKI Testing

To test that the PKI is working, a client without the PKI code will attempt to send its telemetry data to the traffic management system which will have PKI checks in place that make sure that the only connections that get accepted are those that have been properly verified using PKI.

6.5 Multiple Drone Simulation

The final test for this project is to see if the traffic management system will support communications with more than one drone. This is important to test as it will prove that the system can be scaled up to a system that can be used in a real world scenario. This was achieved by using forks to split the traffic management system into waiting for a connection and dealing with an existing connection. The pseudocode can be described as:

```
fork process
if child then print drone name and drone data
if parent then close socket
```

The drone simulation also had to be updated to also send the name of the drone with it so that the traffic management system can identify what drone sent it the telemetry data. Using these three test conditions it was possible to test if the system had secure communication and was able to be scaled up to a larger system.

7 Results

This section will discuss the results of the testing methods described in the previous section, communication with an without TLS, using PKI verification

and not, and connecting to the traffic management system with more than one drone.

7.1 Wireshark Results

Figure 1 and Figure 2 show the result of using Wireshark to inspect the packets sent between the drone simulation and traffic management system virtual machines. Figure 1 shows the result of sending unencrypted messages between the two parties. Wireshark shows that the protocol is TCP and in the bottom of the image, the altitude, speed, longitude and latitude for that particular transmission can be seen. This is a vulnerability as in a real world scenario this data may contain more sensitive or vital information that an attacker could potentially exploit. Looking at Figure 2 it can be seen that the protocol being used is TLSv1.3 and looking at the bottom where a message is, the data is encrypted and cannot be easily obtained by a malicious party. This result demonstrates that the usage of TCP does increase the security of messages sent between a drone and a traffic management system.

7.2 PKI Results

While the PKI method was not set up how it should be in a real world scenario by using a validates Certificate Authority rather than generating the certificate, Figure 3 shows what happens when you connect to the traffic management system without having valid certificates. In a normal situation, when the drone simulation connects to the traffic management system, basic telemetry data will be communicated, but in this case, an error saying the traffic management system reset the communication is sent back. This error message occurs when there is an issue validating certificates which is what was being tested. This result demonstrates that the server correctly refused connections from drones or any other device that does not have a valid authorisation.

7.3 Multiple Drone Results

The last test was to see if the the traffic management system could handle multiple drone connections. Fig-

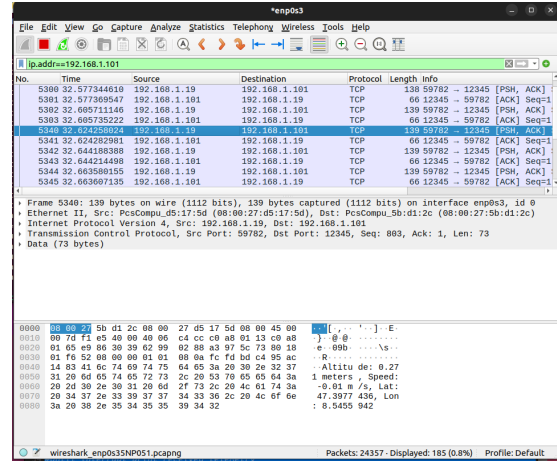


Figure 1: Unencrypted Wireshark Messages

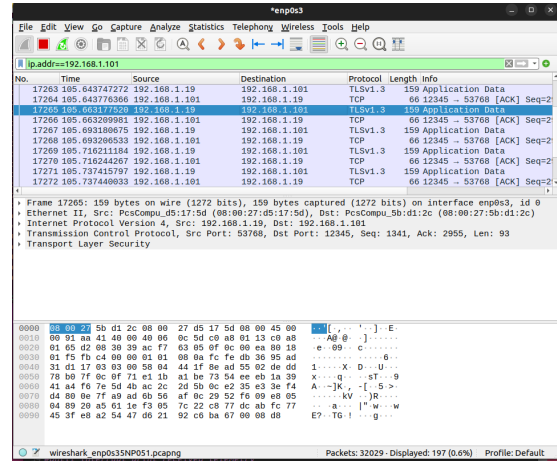


Figure 2: Encrypted Wireshark Messages

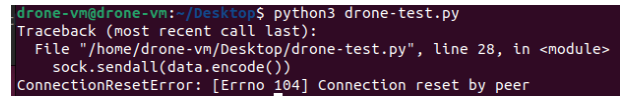


Figure 3: Drone With No PKI Certificate

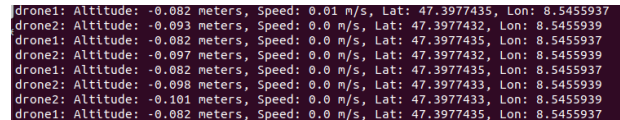


Figure 4: Drone With No PKI Certificate

ure 4 shows the telemetry data that is received by the traffic management system. There are two drones (drone1 and drone2) with each having their own data that is not being overwritten due to the load on the server. This is an important result that demonstrates this systems ability to be scaled into a larger system that can have hundreds or even thousands of drones connecting to the traffic management system.

Overall, the results all show that the system can be used for secure communication and can handle multiple drones being connected to one traffic management system.

8 Conclusion

The purpose of this project was to address the security gap that currently exists in the aviation industry. This project focused specifically on the communication between UAVs (specifically drones) and a traffic management system and attempted to create a method of secure communication using Public Key Infrastructure (PKI) and Transport Layer Security (TLS). After researching how existing systems could be brought together to create secure communication, a result was achieved that showed a possible solution to this gap.

This project successfully used TLS to encrypt messages between a simulated drone and a traffic management system and used PKI to ensure that only valid, authorised drones communicate with the traffic management system. Encrypted and unencrypted messages were analysed using Wireshark, which showed that the usage of TLS provided added security to otherwise completely insecure messages. Another system that was not verified by a certificate authority was used to attempt to connect to the traffic management system where the result showed that the traffic management system would reject messages from unauthorised sources. Finally the project demonstrated the ability of this system to handle more than one connection to the traffic management system by having a two simulated drones connect to the system simultaneously.

These three results show that this system could be used to apply measures to make communication in

aviation more secure, specifically in UAV communications over the internet. This work can be used to develop a more capable system that can be used in real world scenarios as this system has some limitations that must be addressed in future work.

A limitation of this project is that it focused solely on the communication between a simulated drone and a traffic management system and has no other compelling features other than proof that secure communication can be achieved. In this project, specifically in the traffic management system, there are no features that exist in actual traffic management systems such as no-fly zones, verifying flight plans, managing routes for collisions and other important features. While a controller for the simulated drone was omitted as the same secure communication methods can be used, in a real world scenario the traffic management system would need to communicate with a controller for its missing features such as flight plans and no fly zones.

While there is a need for future work for this project, the results show that secure communication in aviation is possible and with new technologies rapidly appearing, the threats of not having secure communication can have disastrous consequences.

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